

VIA.STACK // Chemical Kinetics Study Notes

SAMPLE PREVIEW

A_0001 // KINETICS

Catalysis & Activation Energy

LOGIC: A catalyst lowers the activation energy by providing an alternative reaction pathway with a lower energy barrier. Enthalpy is a state function that depends only on the energy of initial and final states, so ΔH remains unchanged. Therefore, catalysts increase the reaction rate without affecting the net thermodynamic energy change.

A_0006 // KINETICS

Second-Order Kinetics

LOGIC: The units of the rate constant are derived from the rate law where $\text{Rate} = k[A]^2$. The rate has units of $\text{mol}/(\text{L s})$ while the concentration squared has units of mol^2/L^2 . Dividing rate by concentration squared results in $\text{L}/(\text{mol s})$.

A_0015 // KINETICS

Arrhenius Temperature Dependency

LOGIC: Activation energy is a characteristic property of a reaction reflecting the energy barrier of the transition state. While temperature increases the fraction of molecules with enough energy to cross the barrier, it does not alter the height of the barrier itself. Therefore, E_a is independent of temperature changes in the standard Arrhenius model.
